

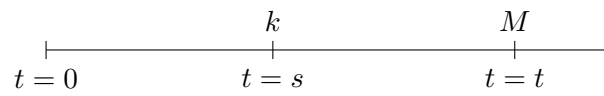
Problem 4.20

Suppose that an account earns simple interest with annual interest rate of i . If an investment of k is made at time s years, what is the accumulated value at time $t > s$ years?

Solution

For simple interest, the accumulation function is

$$a(t) = 1 + it.$$



If the deposit of amount k is made at time s , then from time s to time t it accumulates for $t - s$ years. Therefore its accumulated value at time t is

$$M = k(1 + i(t - s)).$$

So the accumulated value at time $t > s$ of an investment of k made at time s is

$$\boxed{M = k(1 + i(t - s))}.$$

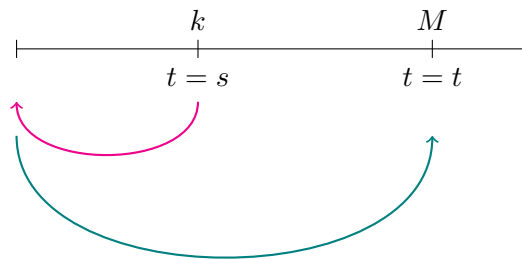
Another equivalent way to see this is to move the deposit back to time 0. The amount at time 0 equivalent to k invested at time s is

$$\frac{k}{1 + is}.$$

Accumulating this amount to time t gives

$$M = \frac{k}{1 + is}(1 + it).$$

Visual explanation of the exercise



Since

$$\frac{1 + it}{1 + is} = 1 + i(t - s)$$

under simple interest, this agrees with the formula above.